20261008 Quiz

for (MATH102) Calculus II

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Problem 2.45. Consider the polar coordinate mapping

$$\mathbf{F}(r,\theta) = (r\cos\theta, r\sin\theta), \qquad r \geq 0, \quad 0 \leq \theta \leq 2\pi.$$

Sketch a region in the (r,θ)-plane that corresponds to the upper half-plane $y > 0$.

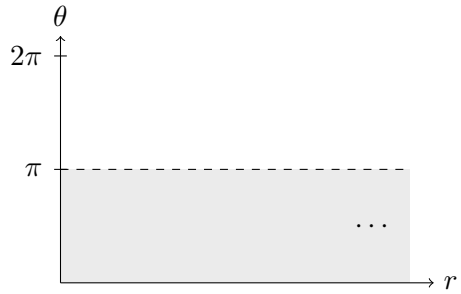
Solution. Since $y = r\sin\theta$, the condition $y > 0$ is equivalent, in the standard polar coordinate range, to

$$r > 0, \qquad 0 < \theta < \pi.$$

Thus the required region is

$$\boxed{\{(r,\theta) \in \mathbb{R}^2 \mid r > 0,\ 0 < \theta < \pi\}}.$$

A sketch in the (r,θ)-plane is shown below.



□

Problem 2.46. Sketch the region in the (x,y)-plane that corresponds to the polar coordinate rectangle

$$1 \leq r \leq 2, \qquad 0 \leq \theta \leq \pi.$$

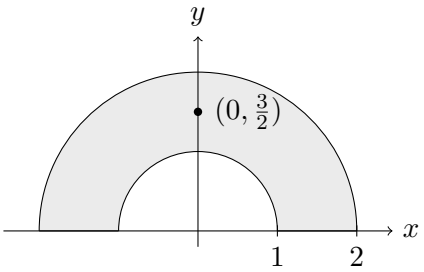
Find the polar coordinates of the point $(x,y) = (0, 1.5)$.

Solution. The condition $1 \le r \le 2$ places the points between the circles of radii 1 and 2, while $0 \le \theta \le \pi$ restricts them to the upper half-plane.

Hence the image is

$$\boxed{\{(x,y) \in \mathbb{R}^2 \mid y \ge 0,\ 1 \le x^2 + y^2 \le 4\}},$$

the closed upper half-annulus shown below.



For $(x, y) = (0, 1.5)$,

$$r = \sqrt{x^2 + y^2} = \frac{3}{2}, \qquad \theta = \frac{\pi}{2}.$$

Therefore its polar coordinates are

$$\boxed{(r, \theta) = \left(\frac{3}{2}, \frac{\pi}{2}\right)}.$$

□

Problem 2.47. Use equations and inequalities to describe the following sets in polar coordinates.

(a) The open unit disk $x^2 + y^2 < 1$.

(b) The first quadrant $x > 0$, $y > 0$.

Solution. Using $x^2 + y^2 = r^2$ and the standard ranges $r \ge 0$ and $0 \le \theta \le 2\pi$,

(a) The open unit disk is described by

$$\boxed{0 \le r < 1, \qquad 0 \le \theta \le 2\pi}.$$

(b) In the first quadrant both $x = r\cos\theta$ and $y = r\sin\theta$ are positive. Thus

$$\boxed{r > 0, \qquad 0 < \theta < \frac{\pi}{2}}.$$

□

Problem 2.49. Let $0 < b < a$ and consider the set of points in the plane whose polar coordinates r, θ satisfy

$$r = \frac{1}{a + b\cos\theta}.$$

(a) Show that the x, y coordinates of the points satisfy

$$1 = a\sqrt{x^2 + y^2} + bx.$$

(b) Show that the equation

$$1 = a\sqrt{x^2 + y^2} + bx$$

is the equation of an ellipse by completing a square and getting the equation into the form

$$1 = \frac{(x - \alpha)^2}{c^2} + \frac{(y - \beta)^2}{d^2}.$$

Solution.

(a) Multiplying the polar equation by $a + b\cos\theta$ gives

$$1 = ar + br\cos\theta.$$

Since

$$r = \sqrt{x^2 + y^2}, \qquad r\cos\theta = x,$$

we obtain

$$\boxed{1 = a\sqrt{x^2 + y^2} + bx}.$$

(b) From the equation in part (a),

$$1 - bx = a\sqrt{x^2 + y^2}.$$

Squaring and rearranging gives

$$(a^2 - b^2)x^2 + 2bx + a^2y^2 = 1.$$

Since $0 < b < a$, we have $a^2 - b^2 > 0$. Completing the square in x gives

$$(a^2 - b^2)\left(x + \frac{b}{a^2 - b^2}\right)^2 + a^2y^2 = \frac{a^2}{a^2 - b^2}.$$

Therefore

$$\boxed{1 = \frac{\left(x + \dfrac{b}{a^2 - b^2}\right)^2}{\dfrac{a^2}{(a^2 - b^2)^2}} + \frac{y^2}{\dfrac{1}{a^2 - b^2}}}.$$

This is an ellipse with

$$\alpha = -\frac{b}{a^2 - b^2}, \qquad \beta = 0, \qquad c = \frac{a}{a^2 - b^2}, \qquad d = \frac{1}{\sqrt{a^2 - b^2}}.$$

Conversely, on this ellipse,

$$x \le \alpha + c = \frac{1}{a + b}, \qquad 1 - bx \ge \frac{a}{a + b} > 0.$$

Thus the squaring step is reversible, and the ellipse is exactly the original set.

□

Problem 3.2. Let

$$f(x, y) = x^2 + 3y.$$

Find the linear approximation of $f(x, y)$ near $(2, 4)$, and use it to estimate $f(2.01, 4.03)$.

Solution. We have

$$f(2, 4) = 16, \qquad f_x(x, y) = 2x, \qquad f_y(x, y) = 3.$$

Thus

$$f_x(2, 4) = 4, \qquad f_y(2, 4) = 3,$$

and the linear approximation at $(2, 4)$ is

$$\boxed{L(x, y) = 16 + 4(x - 2) + 3(y - 4)}.$$

Using this approximation,

$$f(2.01, 4.03) \approx L(2.01, 4.03) = 16 + 4(0.01) + 3(0.03) = 16.13.$$

Thus the requested estimate is

$$\boxed{f(2.01, 4.03) \approx 16.13}.$$

□

Problem 3.3. Find the indicated partial derivatives of the functions.

(a) $f_x(x, y)$ and $f_y(2, 0)$ if

$$f(x, y) = e^{-x^2-y^2}.$$

(b)

$$\frac{\partial}{\partial y}\left(xe^y + ye^x\right).$$

(c)

$$\frac{\partial}{\partial x}\left(\cos(xy) + \frac{\partial}{\partial y}\sin(xy)\right).$$

Solution.

(a) Differentiating with respect to x gives

$$\boxed{f_x(x, y) = -2xe^{-x^2-y^2}}.$$

Also,

$$f_y(x, y) = -2ye^{-x^2-y^2},$$

so

$$\boxed{f_y(2, 0) = 0}.$$

(b) Holding x fixed,

$$\boxed{\frac{\partial}{\partial y}\left(xe^y + ye^x\right) = xe^y + e^x}.$$

(c) First,

$$\frac{\partial}{\partial y}\sin(xy) = x\cos(xy).$$

Therefore

$$\begin{aligned}\frac{\partial}{\partial x}\left(\cos(xy) + \frac{\partial}{\partial y}\sin(xy)\right) &= \frac{\partial}{\partial x}\left(\cos(xy) + x\cos(xy)\right) \\ &= \boxed{\cos(xy) - y\sin(xy) - xy\sin(xy)}.\end{aligned}$$

□

Problem 3.6. Consider the linear function

$$\ell(x, y, z) = ax + by + cz.$$

Show that $\nabla\ell$ is constant. Show that when a, b, c are not all zero, $\nabla\ell$ is normal to the level set $\ell = 0$.

Solution. The partial derivatives are

$$\ell_x = a, \qquad \ell_y = b, \qquad \ell_z = c,$$

so

$$\boxed{\nabla \ell = (a, b, c)},$$

which is constant.

Now let $\mathbf{x} = (x, y, z)$ lie in the level set $\ell = 0$. Then

$$\nabla \ell \cdot \mathbf{x} = (a, b, c) \cdot (x, y, z) = ax + by + cz = 0.$$

Thus the level set $\ell = 0$ is the plane through the origin orthogonal to $\nabla \ell$. Since a, b, c are not all zero, $\nabla \ell \neq \mathbf{0}$, so $\nabla \ell$ is a normal vector to the level set.

□

Problem 3.11. Define functions $f, g \colon \mathbb{R}^2 \to \mathbb{R}$ by

$$f(x, y) = \cos(x + y), \qquad g(x, y) = \sin(2x - y).$$

Find the gradients ∇f and ∇g, and show that

$$f_x - f_y = 0, \qquad g_x + 2g_y = 0.$$

Solution. For f,

$$f_x(x, y) = -\sin(x + y), \qquad f_y(x, y) = -\sin(x + y),$$

so

$$\boxed{\nabla f(x, y) = (-\sin(x + y), -\sin(x + y))}.$$

Hence

$$f_x - f_y = -\sin(x + y) + \sin(x + y) = 0.$$

For g,

$$g_x(x, y) = 2\cos(2x - y), \qquad g_y(x, y) = -\cos(2x - y),$$

so

$$\boxed{\nabla g(x, y) = (2\cos(2x - y), -\cos(2x - y))}.$$

Hence

$$g_x + 2g_y = 2\cos(2x - y) - 2\cos(2x - y) = 0.$$

□